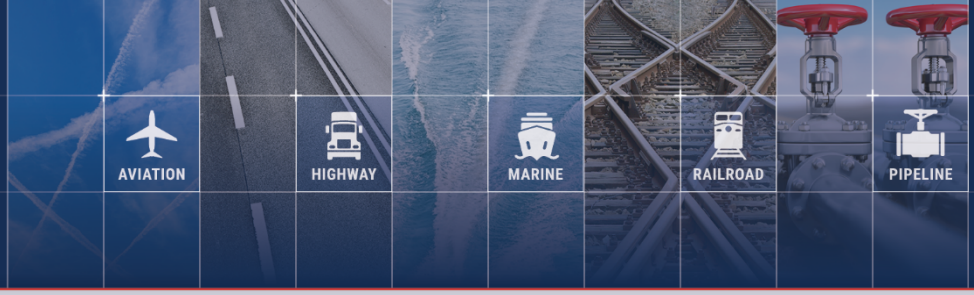

Aviation Investigation Final Report

Location:	Melbourne, Florida	**Accident Number:**	ERA25LA084
Date & Time:	December 26, 2024, 18:57 Local	**Registration:**	N76NN
Aircraft:	Beech BE-76	**Aircraft Damage:**	Substantial
Defining Event:	Fuel exhaustion	**Injuries:**	1 Serious, 1 Minor
Flight Conducted Under:	Part 91: General aviation - Instructional		

Analysis

The pilot and the pilot-rated passenger were flying the multi-engine airplane on a long cross-country trip. At an intermediate stop, the pilot had the airplane serviced to a total fuel quantity of 60 gallons and estimated that the fuel consumption for the planned 3-hour return flight would be slightly more than 9 gallons per hour, per engine. What he did not realize was that the total fuel consumption displayed to him by his flight planning application was only accounting for 9 gallons per hour total, or about half of what it should have been.

About 3 hours after departure, the right engine began to sputter and subsequently lost total power. After declaring a fuel emergency to air traffic control, the left engine also lost power. The pilot and pilot-rated passenger determined that they would not be able to glide the airplane to a nearby airport and they began to scan for an alternate landing location. They attempted to land in a large parking lot, but the airplane struck power lines before contacting the ground. The airplane came to rest upright on the main landing gear. The left wing was separated outboard of the left engine, and the fuselage, right wing, and empennage sustained substantial damage. A small postimpact fire ignited near the portion of the left wing that separated from the fuselage and was quickly extinguished after the accident. Postaccident examination of the airplane revealed that less than 1 gallon of fuel remained in the right wing fuel tank and no fuel was present in what remained of the left wing fuel tank. There was no odor of fuel at the accident site. Given this information, it is likely that the loss of engine power to both engines was due to the pilot exhausting the airplane's usable fuel supply.

Probable Cause and Findings

The National Transportation Safety Board determines the probable cause(s) of this accident to be:

The pilot’s inadequate preflight fuel planning which resulted in a total loss of engine power due to fuel exhaustion and subsequent forced landing.

Findings

Personnel issues	Decision making/judgment - Pilot
Personnel issues	Decision making/judgment - Flight crew

Factual Information

History of Flight

Enroute	Fuel exhaustion (Defining event)
Landing	Collision with terr/obj (non-CFIT)

Pilot Information

Certificate:	Flight instructor	**Age:**	28,Male
Airplane Rating(s):	Single-engine land	**Seat Occupied:**	Left
Other Aircraft Rating(s):	None	**Restraint Used:**	3-point
Instrument Rating(s):	Airplane	**Second Pilot Present:**	Yes
Instructor Rating(s):	Airplane single-engine	**Toxicology Performed:**	
Medical Certification:	Class 1 Without waivers/limitations	**Last FAA Medical Exam:**	September 14, 2022
Occupational Pilot:	Yes	**Last Flight Review or Equivalent:**	October 19, 2023
Flight Time:	1226 hours (Total, all aircraft), 18 hours (Total, this make and model), 1147 hours (Pilot In Command, all aircraft), 175 hours (Last 90 days, all aircraft), 40 hours (Last 30 days, all aircraft)		

Pilot-rated passenger Information

Certificate:	Flight instructor	**Age:**	24,Male
Airplane Rating(s):	Single-engine land; Single-engine sea; Multi-engine land	**Seat Occupied:**	Right
Other Aircraft Rating(s):	None	**Restraint Used:**	3-point
Instrument Rating(s):	Airplane	**Second Pilot Present:**	Yes
Instructor Rating(s):	Airplane multi-engine; Airplane single-engine	**Toxicology Performed:**	
Medical Certification:	Class 1 With waivers/limitations	**Last FAA Medical Exam:**	October 18, 2024
Occupational Pilot:	Yes	**Last Flight Review or Equivalent:**	May 26, 2000
Flight Time:	940 hours (Total, all aircraft), 45 hours (Total, this make and model), 851 hours (Pilot In Command, all aircraft), 196 hours (Last 90 days, all aircraft), 45 hours (Last 30 days, all aircraft)		

Aircraft and Owner/Operator Information

Aircraft Make:	Beech	**Registration:**	N76NN
Model/Series:	BE-76 NO SERIES	**Aircraft Category:**	Airplane
Year of Manufacture:	1979	**Amateur Built:**	
Airworthiness Certificate:	Normal	**Serial Number:**	ME-158
Landing Gear Type:	Retractable - Tricycle	**Seats:**	4
Date/Type of Last Inspection:	October 24, 2024 Annual	**Certified Max Gross Wt.:**	3900 lbs
Time Since Last Inspection:	39 Hrs	**Engines:**	2
Airframe Total Time:	5593.2 Hrs as of last inspection	**Engine Manufacturer:**	
ELT:	C126 installed, activated, did not aid in locating accident	**Engine Model/Series:**	
Registered Owner:	MELBOURNE FLIGHT TRAINING LLC	**Rated Power:**	
Operator:	MELBOURNE FLIGHT TRAINING LLC	**Operating Certificate(s) Held:**	None

Meteorological Information and Flight Plan

Conditions at Accident Site:	Visual (VMC)	**Condition of Light:**	Night
Observation Facility, Elevation:	MLB,22 ft msl	**Distance from Accident Site:**	3 Nautical Miles
Observation Time:	18:53 Local	**Direction from Accident Site:**	104°
Lowest Cloud Condition:	Clear	**Visibility**	10 miles
Lowest Ceiling:	Overcast / 4500 ft AGL	**Visibility (RVR):**	
Wind Speed/Gusts:	6 knots / None	**Turbulence Type Forecast/Actual:**	None / None
Wind Direction:	30°	**Turbulence Severity Forecast/Actual:**	N/A / N/A
Altimeter Setting:	30.11 inches Hg	**Temperature/Dew Point:**	19°C / 17°C
Precipitation and Obscuration:	No Obscuration; No Precipitation		
Departure Point:	Rock Hill, SC (UZA)	**Type of Flight Plan Filed:**	VFR
Destination:	Melbourne, FL (MLB)	**Type of Clearance:**	VFR
Departure Time:	15:55 Local	**Type of Airspace:**	Class D

Airport Information

Airport:	MELBOURNE ORLANDO INTL MLB	**Runway Surface Type:**	
Airport Elevation:	33 ft msl	**Runway Surface Condition:**	Dry
Runway Used:		**IFR Approach:**	None
Runway Length/Width:		**VFR Approach/Landing:**	Forced landing

Wreckage and Impact Information

Crew Injuries:	1 Serious, 1 Minor	**Aircraft Damage:**	Substantial
Passenger Injuries:	N/A	**Aircraft Fire:**	On-ground
Ground Injuries:	N/A	**Aircraft Explosion:**	None
Total Injuries:	1 Serious, 1 Minor	**Latitude, Longitude:**	28.115083,-80.706661(est)

Preventing Similar Accidents

Prevent the Preventable with Careful Fuel Management (SA-067)

The Problem

Within fuel-related accidents, fuel exhaustion and fuel starvation continue to be leading causes. From 2011 to 2015, an average of more than 50 accidents per year occurred due to fuel management issues. Fuel exhaustion accounted for 56% of fuel-related accidents while fuel starvation was responsible for 35% of these accidents. Fuel exhaustion is running out of fuel whereas fuel starvation is having fuel onboard that doesn’t reach the engine for reasons such as a blockage, improperly set fuel selector, or water contamination.

Running out of fuel or starving an engine of fuel is highly preventable. An overwhelming majority of our investigations of fuel management accidents—95%—cited personnel issues (such as use of equipment, planning, or experience in the type of aircraft being flown) as

causal or contributing to fuel exhaustion or starvation accidents. Prudent pilot action can eliminate these issues. Less than 5% of investigations cited a failure or malfunction of the fuel system.

What can you do?

- Pilots should know how much fuel they have onboard at all times.
- During preflight inspection, measure or visually confirm the fuel quantity. Do not rely exclusively on fuel gauges.
- Know how much fuel you will need for a given flight.
- Make sure you have a fuel reserve for each flight.
- Know your engine's fuel burn rate and actively monitor the fuel burn rate for the entire time the engine is operating.
- Know your aircraft's fuel system and how it works.
- Review your aircraft's POH and use the appropriate checklists.
- Don't stretch your available fuel supply. Stop and get gas!

See https://www.ntsb.gov/Advocacy/safety-alerts/Documents/SA-067.pdf for additional resources.

The NTSB presents this information to prevent recurrence of similar accidents. Note that this should not be considered guidance from the regulator, nor does this supersede existing FAA Regulations (FARs).

Administrative Information

Investigator In Charge (IIC):	Enders, Ryan
Additional Participating Persons:	Matthew Boucher; FAA/FSDO; Orlando, FL
Original Publish Date:	October 30, 2025
Last Revision Date:	
Investigation Class:	Class 4
Note:	The NTSB did not travel to the scene of this accident.
Investigation Docket:	https://data.ntsb.gov/Docket?ProjectID=199481

The National Transportation Safety Board (NTSB) is an independent federal agency charged by Congress with investigating every civil aviation accident in the United States and significant events in other modes of transportation—railroad, transit, highway, marine, pipeline, and commercial space. We determine the probable causes of the accidents and events we investigate, and issue safety recommendations aimed at preventing future occurrences. In addition, we conduct transportation safety research studies and offer information and other assistance to family members and survivors for each accident or event we investigate. We also serve as the appellate authority for enforcement actions involving aviation and mariner certificates issued by the Federal Aviation Administration (FAA) and US Coast Guard, and we adjudicate appeals of civil penalty actions taken by the FAA.

The NTSB does not assign fault or blame for an accident or incident; rather, as specified by NTSB regulation, "accident/incident investigations are fact-finding proceedings with no formal issues and no adverse parties … and are not conducted for the purpose of determining the rights or liabilities of any person" (Title 49 *Code of Federal Regulations* section 831.4). Assignment of fault or legal liability is not relevant to the NTSB's statutory mission to improve transportation safety by investigating accidents and incidents and issuing safety recommendations. In addition, statutory language prohibits the admission into evidence or use of any part of an NTSB report related to an accident in a civil action for damages resulting from a matter mentioned in the report (Title 49 *United States Code* section 1154(b)). A factual report that may be admissible under 49 *United States Code* section 1154(b) is available here.